

Making Fusion 360 Libraries

Importing Eagle library files from SnapMagic

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Introduction

This document explains the steps used to create electronics libraries in Fusion360 by importing Eable library files (.lbr) from SnapMagic on a Mac. This process assumes that you have already downloaded the appropriate .lbr and associated .STEP files from SnapMagic.

Interface

This section helps clarify the UI terms used in this document.

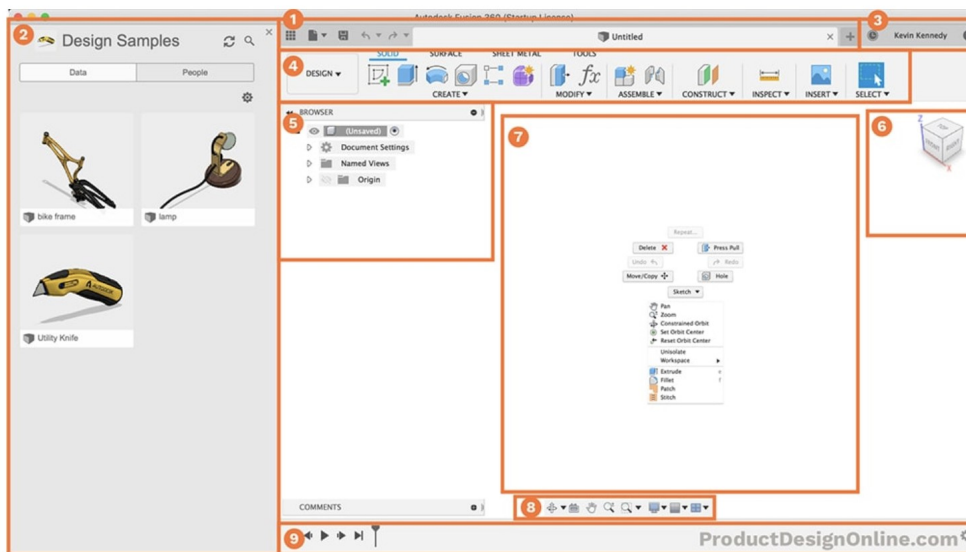


Figure 1

1. Application Bar
2. Data Panel
3. Profile and Help
4. Toolbar
5. Browser
6. ViewCube
7. Canvas and Marking Menu
8. Navigation Bar and Display Settings
9. Timeline

Terminology

An electronics library project is the project “directory” that appears at the top of the data panel of the Fusion360 UI right next to the home icon.

A component appears as a file inside the electronics project. It contains a symbol, a footprint and a 3D model and allows you to manage the mapping between all three items.

A 3D model is a STEP file that is mapped to a component file via the content manager in the electronics component UI.

Process

Step 1: Make a new project to hold your libraries

From the home screen create a new project called ElectronicsLibraries

This makes a project under home

Pin it so it appears at the top of the project list

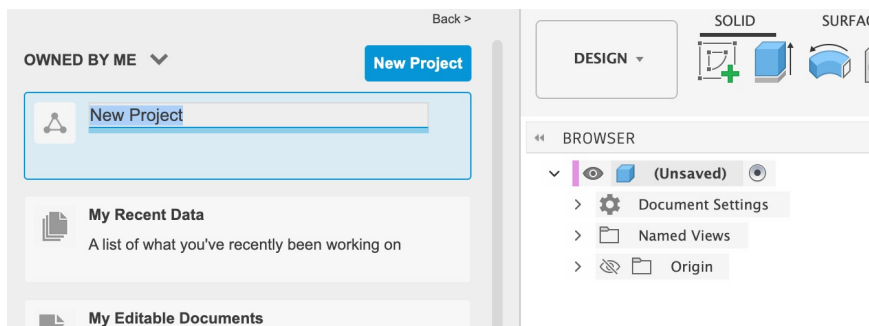


Figure 2

Step 2: Create a new Electronics project

From the tool bar create a new Electronics Library

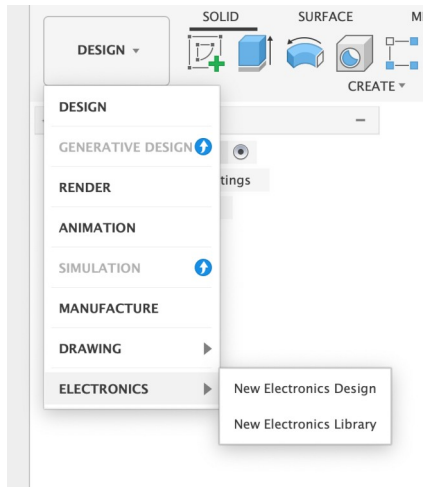


Figure 3

Step 3: Import a SnapMagic library file

Click the “Import Component” button in the center of the “Canvas and Marking Menu” panel.

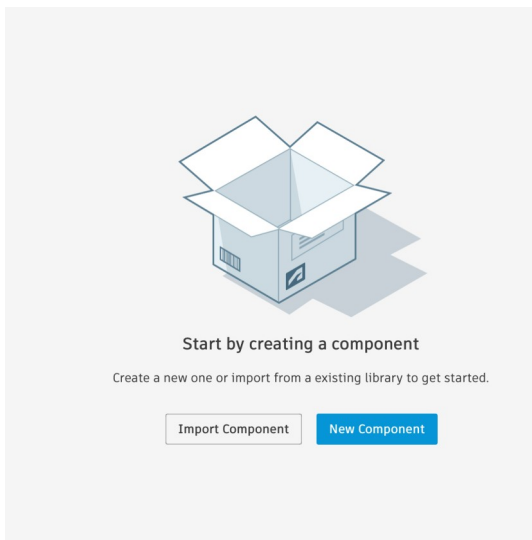


Figure 4

Next, click the “Open Library Manager” button in the bottom left corner of the pop up screen.

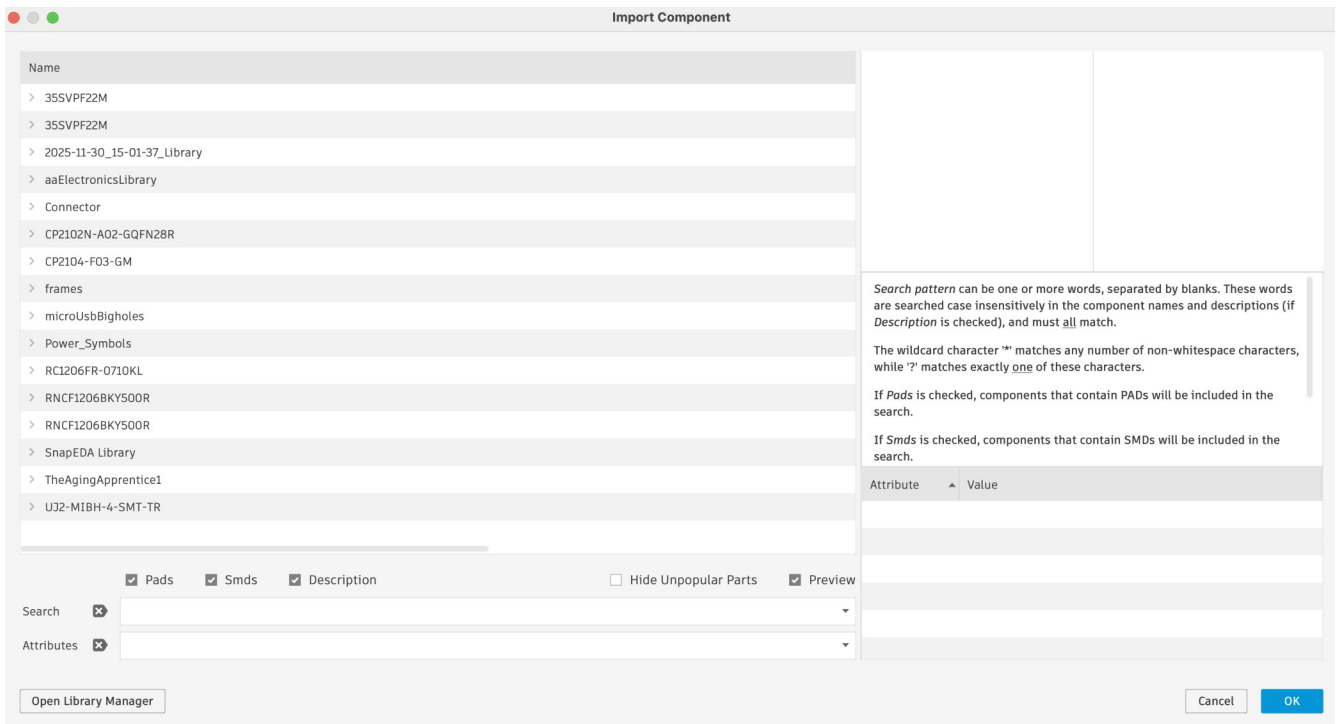


Figure 5

Now activate the Private Libraries tab in the top right corner of the pop up window.

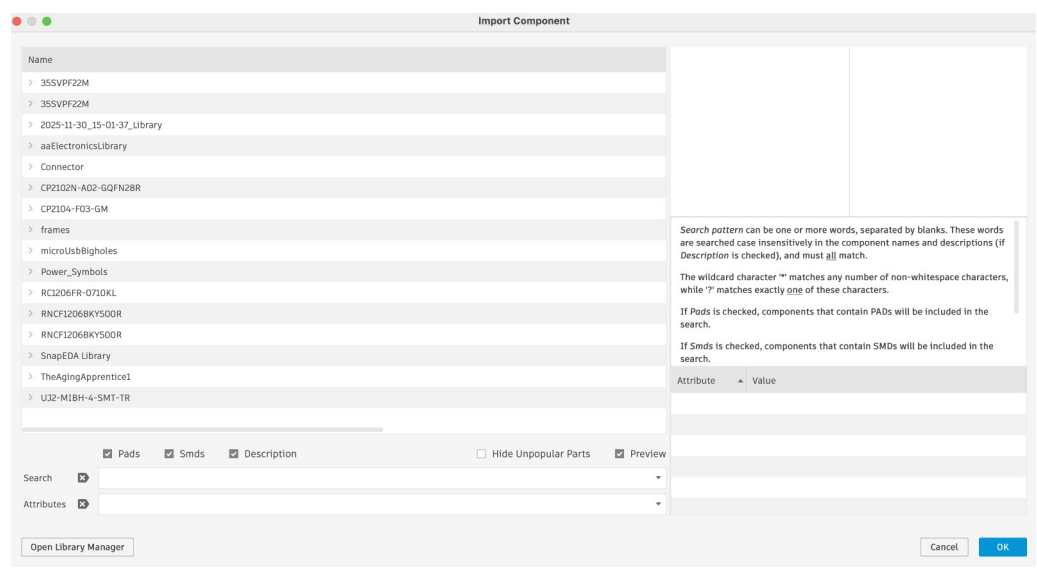


Figure 6

Then click the Local Disk check box

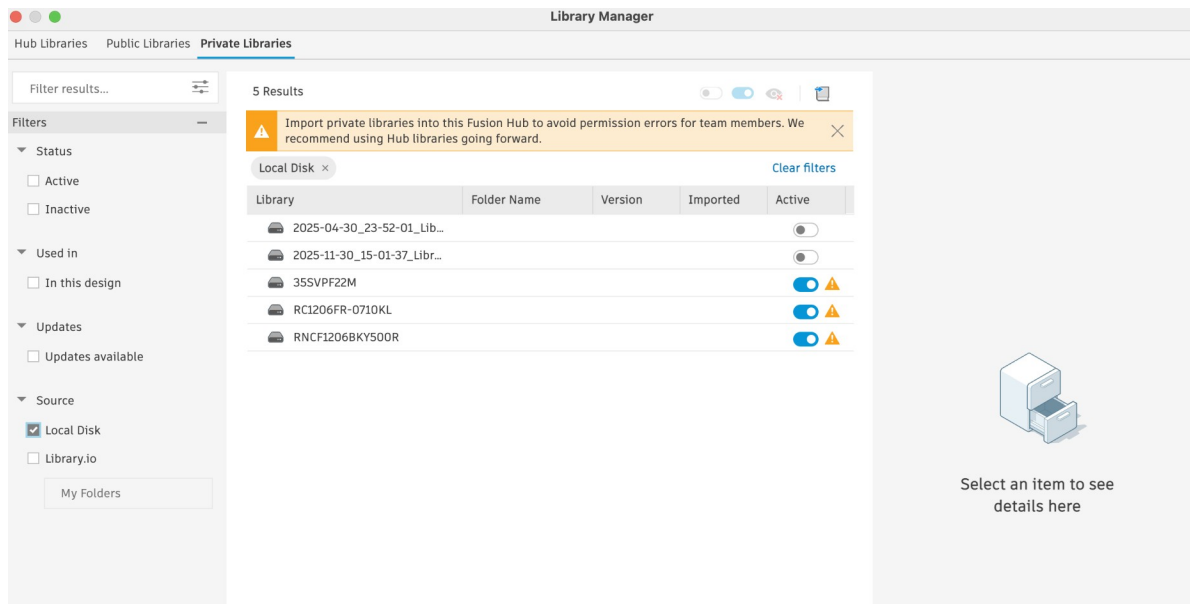


Figure 7

Click the Sync Libraries icon in the upper right corner of the screen

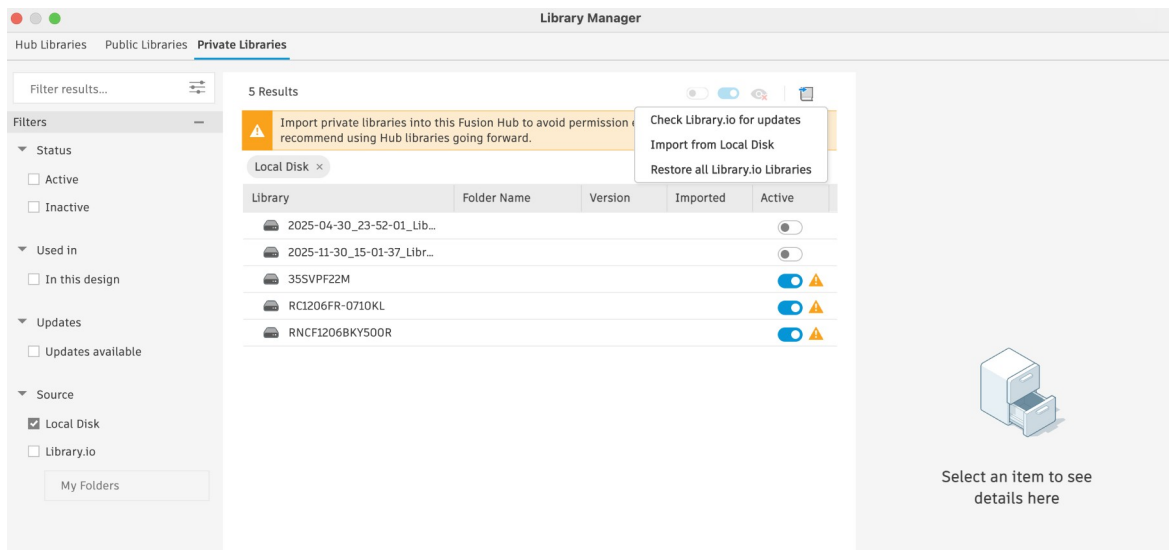


Figure 8

Select the “Import from Local Disk” option

Navigate to the .lbr file and select Open

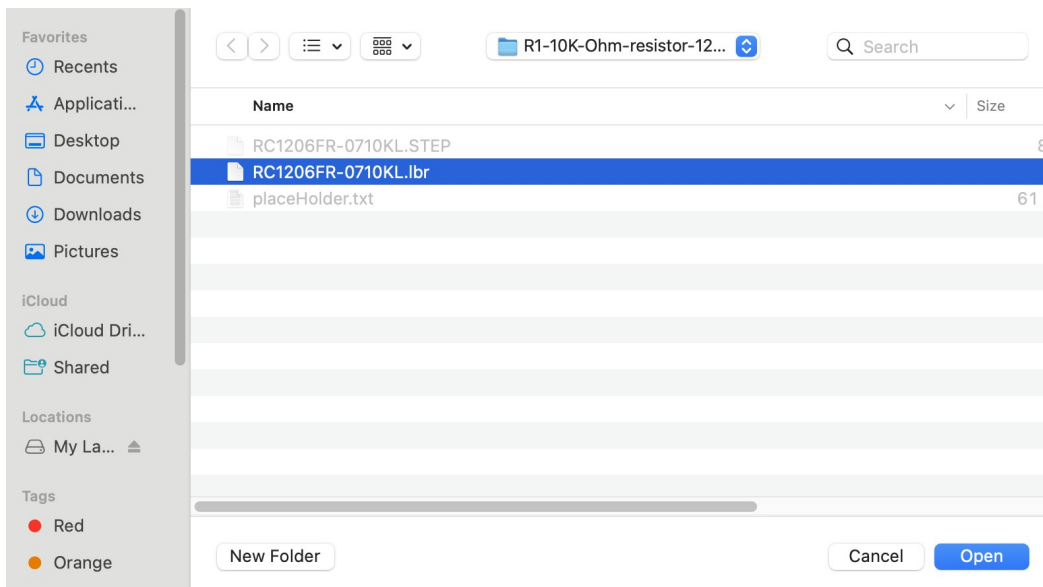


Figure 9

Click import into Fusion Hub button IN the upper right corner of the screen.
Then click the Im[port Into Fusion Hub button (leave the Decativate the local disk library option) checked.

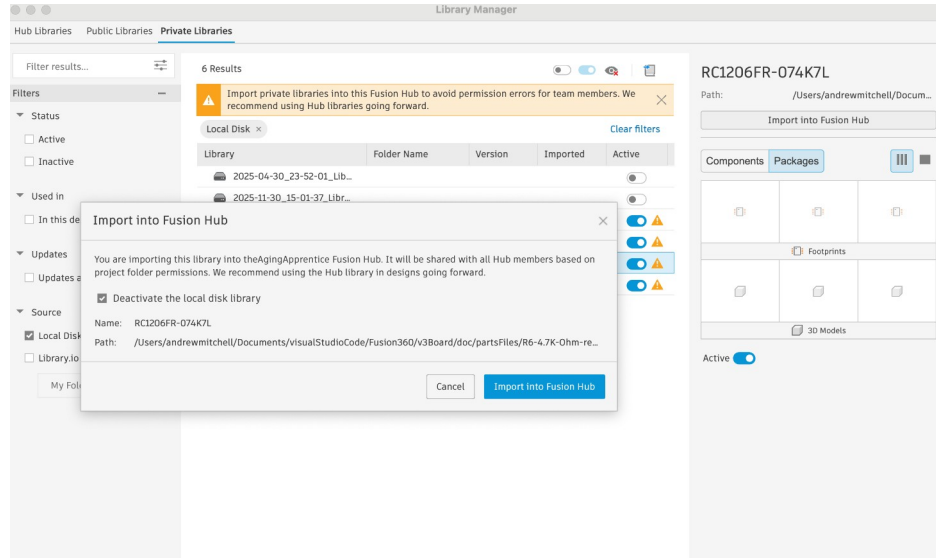


Figure 10

Click Cancel to close the unwanted dialog box and you will see the the component with a footprint and a Symbol. Note that at this point there is no 3D model.

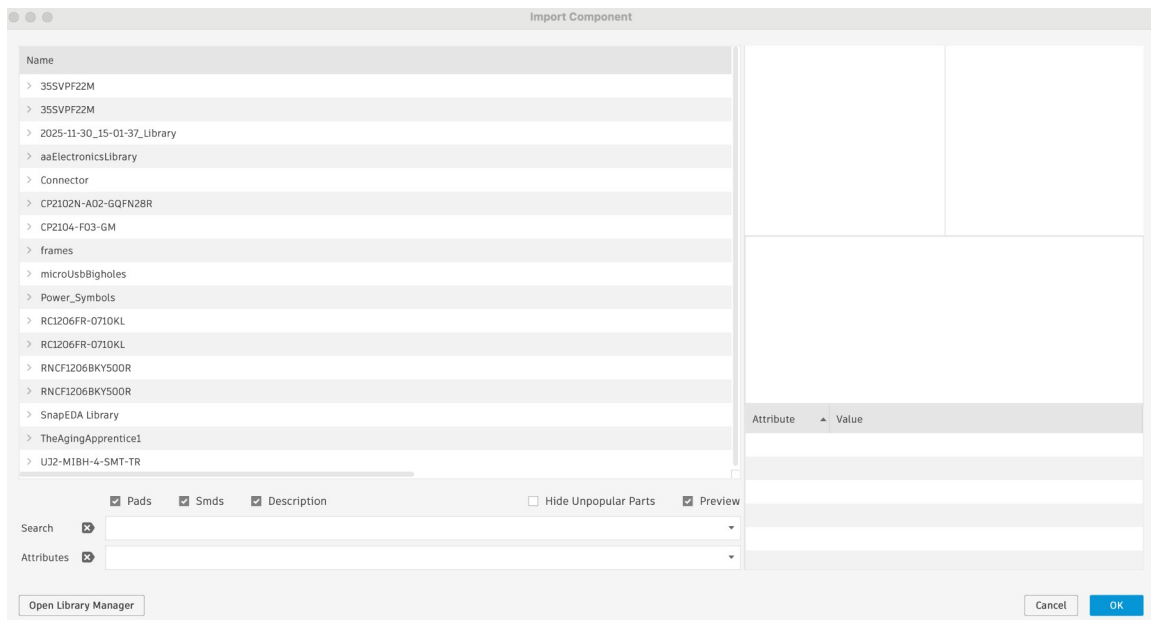


Figure 11

Finally, click the file icon in the applicaiotn bar and select save. Be sure to select the Electrocoics Library project that you created earlier as your location.

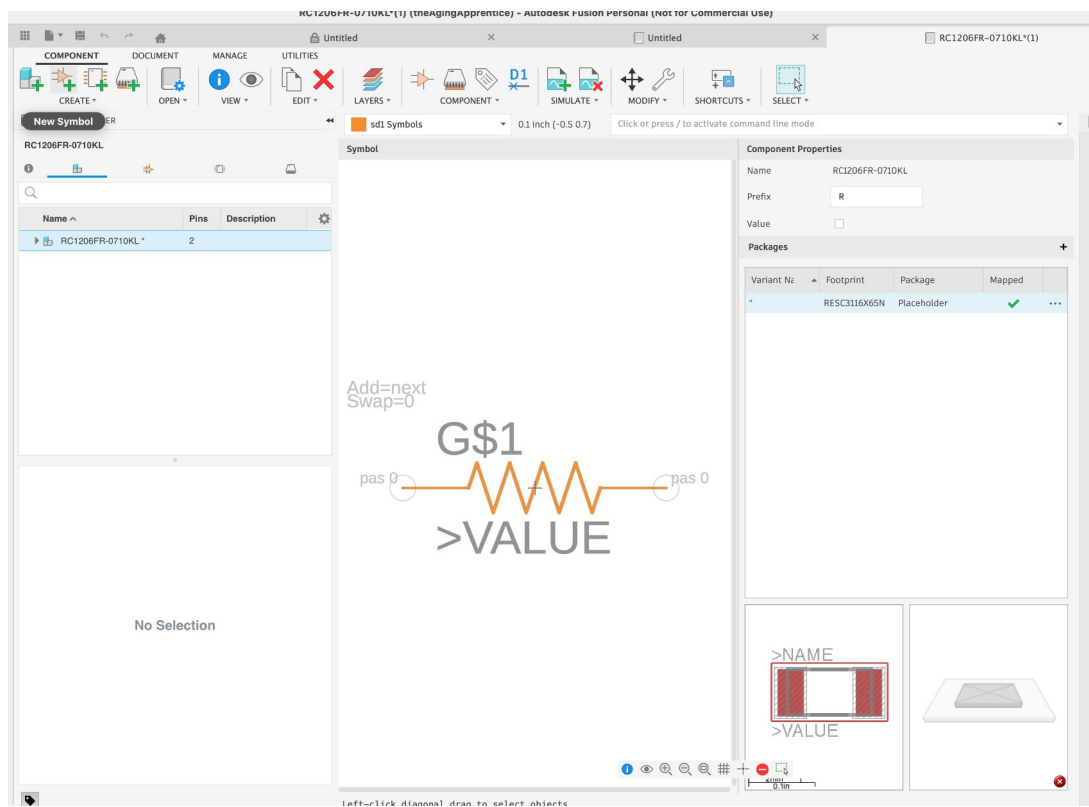


Figure 12

Step 3: Add the 3D model to the project

Click the Upload button in the upper left in the data panel.

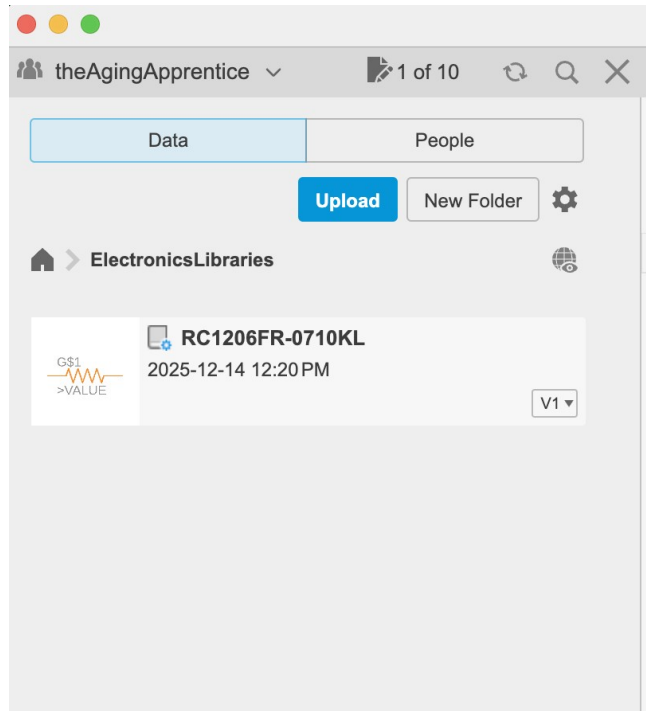


Figure 13

Click the Select button on the upload dialog pop up window

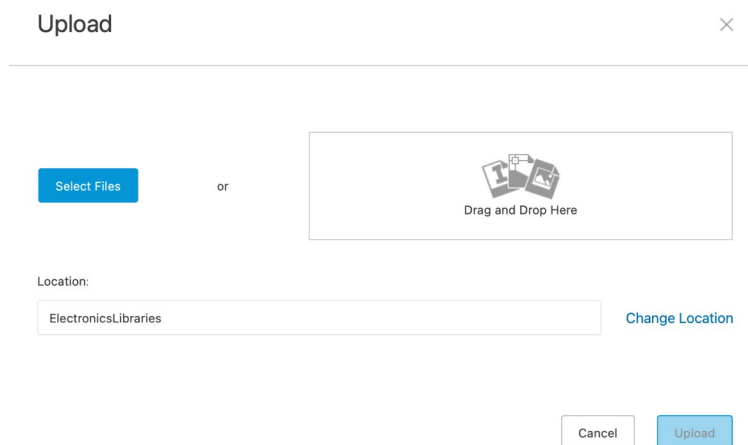


Figure 14

Navigate to the STEP file on your hard drive, select the file and click the Open button.

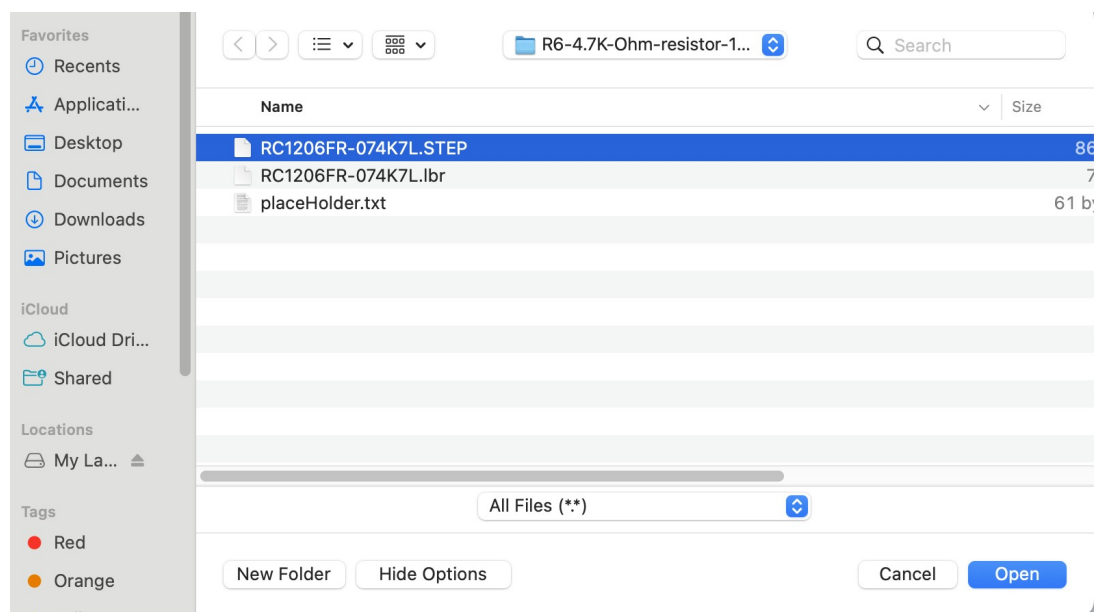


Figure 15

Now click the upload button

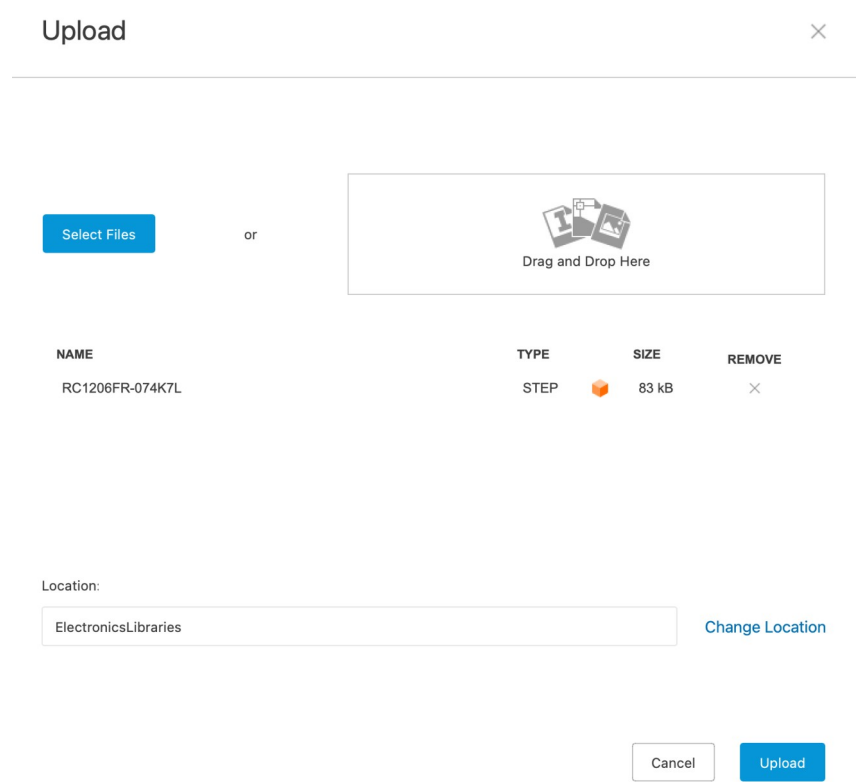


Figure 16

You should now see both the .lbr file (component) and the .STEP file in the data panel.

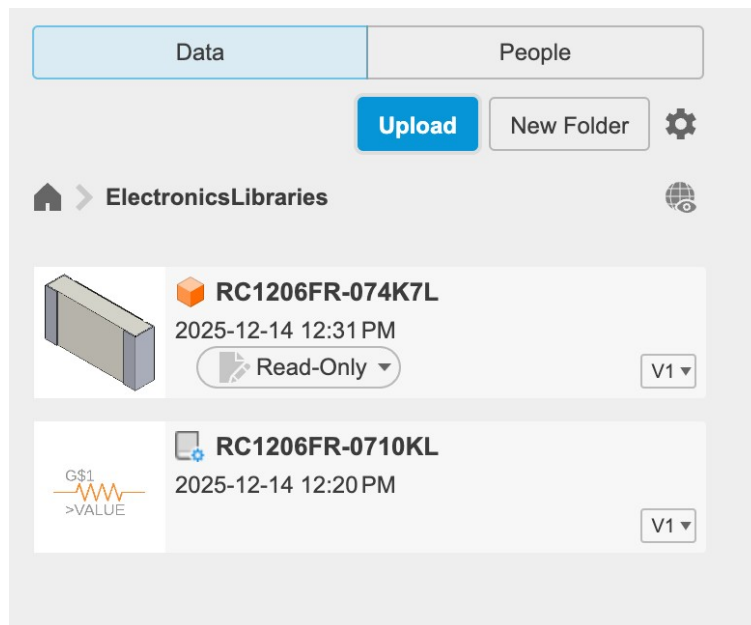


Figure 17

Step 4: Add the 3d model to the component package

Open the target library in the project by double clicking it's file in the data panel. You can expand the component and see that it has a Symbol and a foot print but no Package.

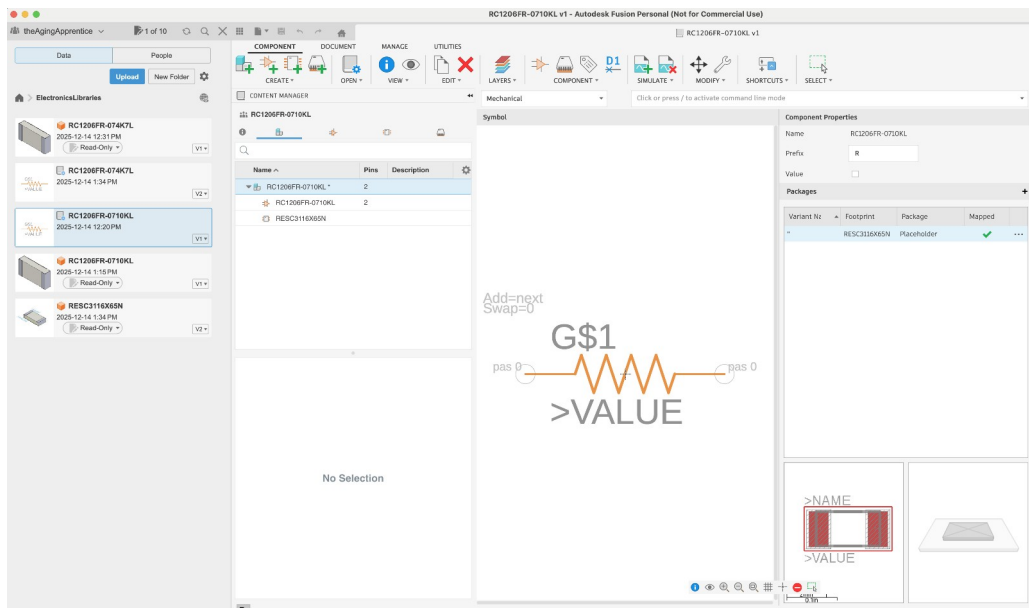


Figure 18

Right click the Footprint in the left panel of the component manager screen.

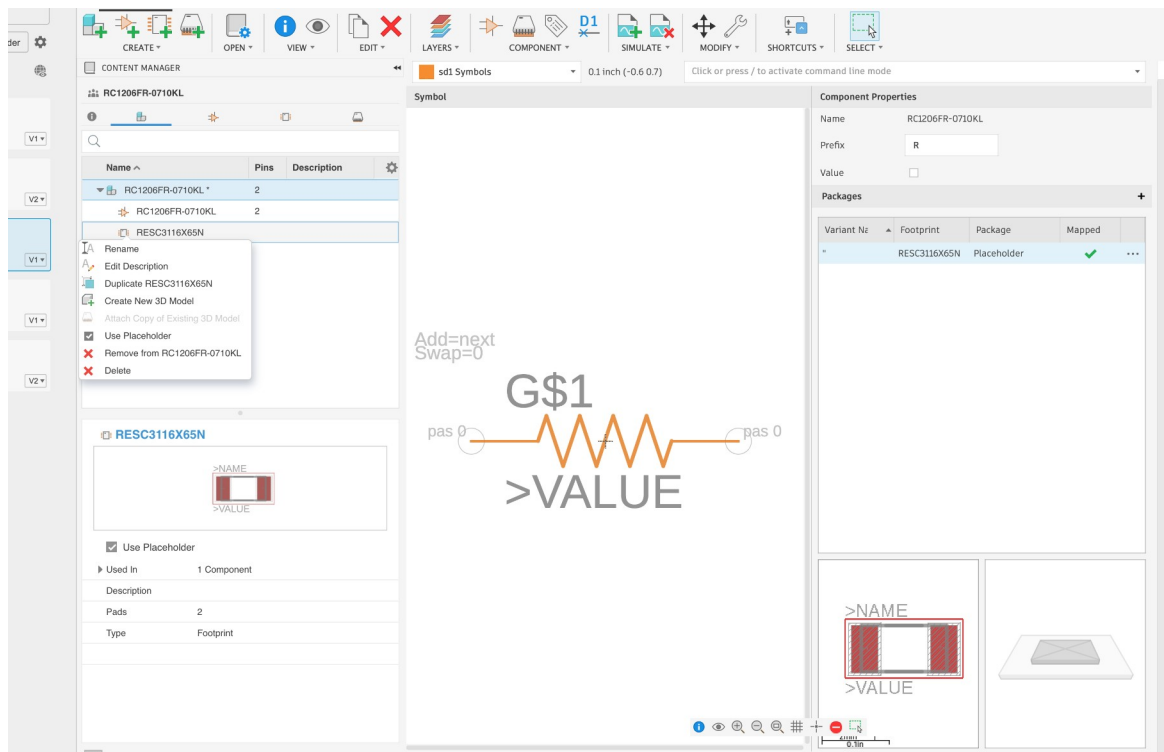


Figure 19

Select the “Create a New 3D model” option from the dropdown list.

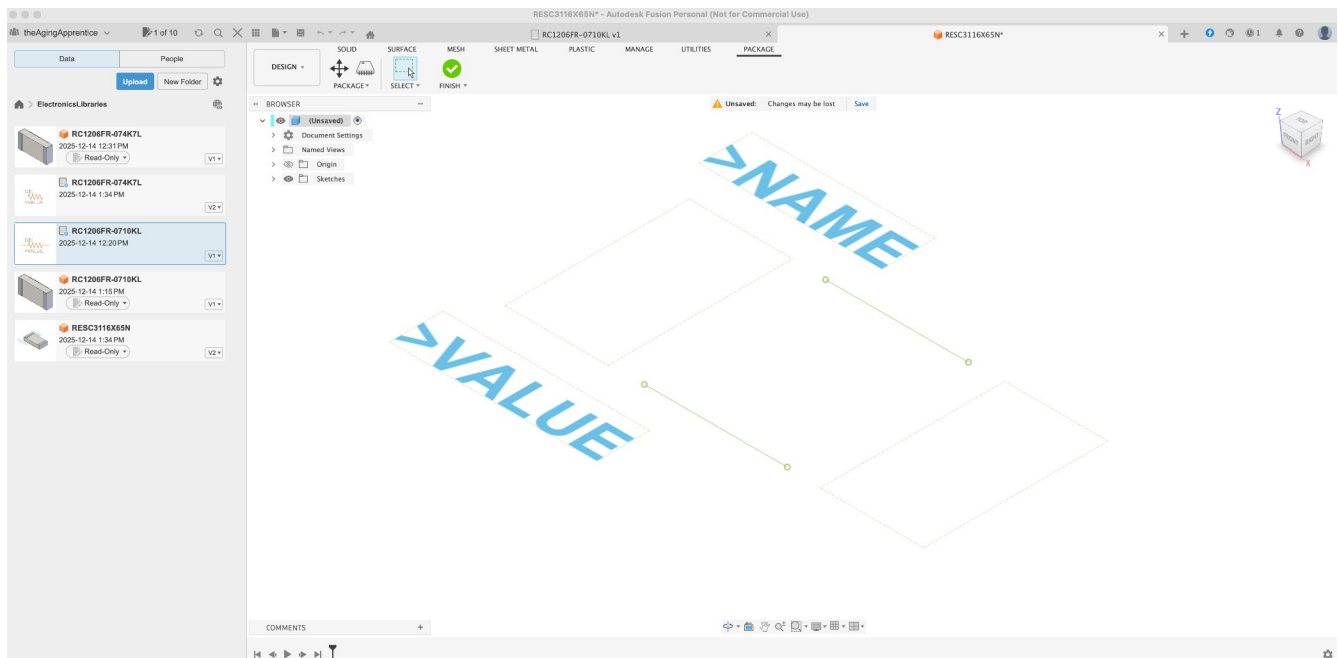


Figure 20

Right click the STEP file (3D model) that you want from the Data Panel and select the “Insert into current design” dropdown menu option.

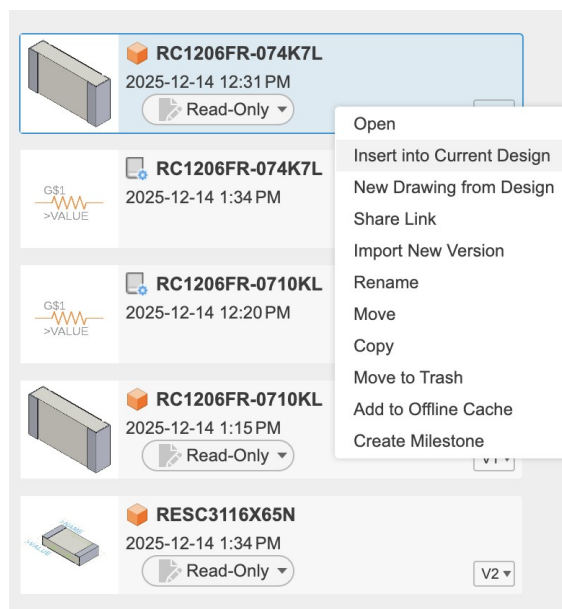


Figure 21

Orient the 3D model dso that it is properly aligned onto the pads (or holes) of the Package.

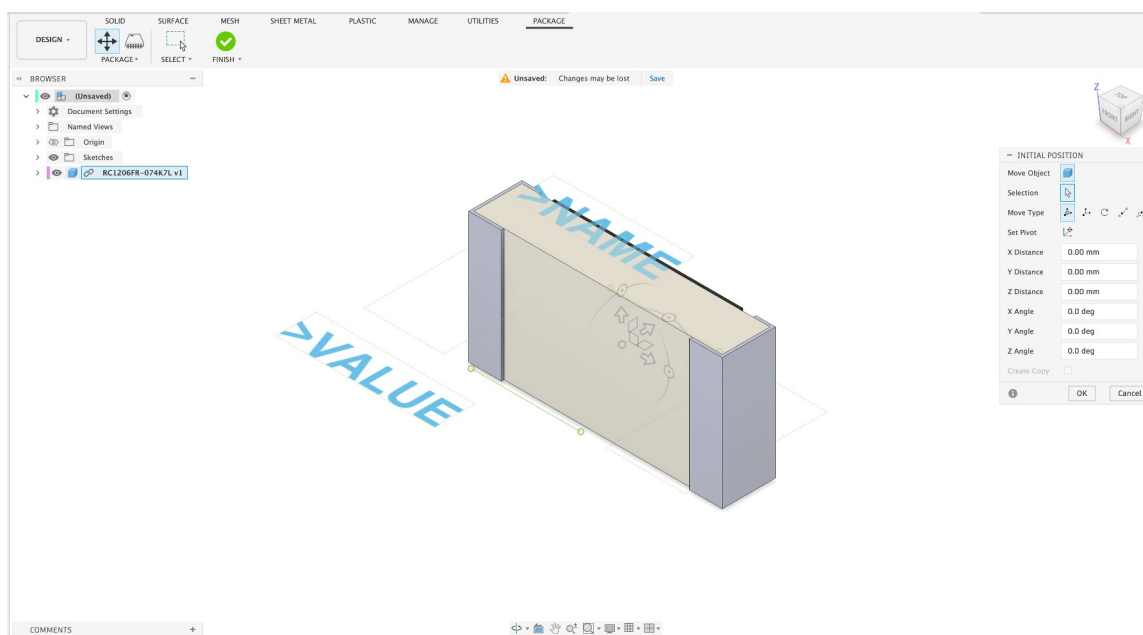


Figure 22

When you are done click OK button in the little navigation pop up panel under the view cube.

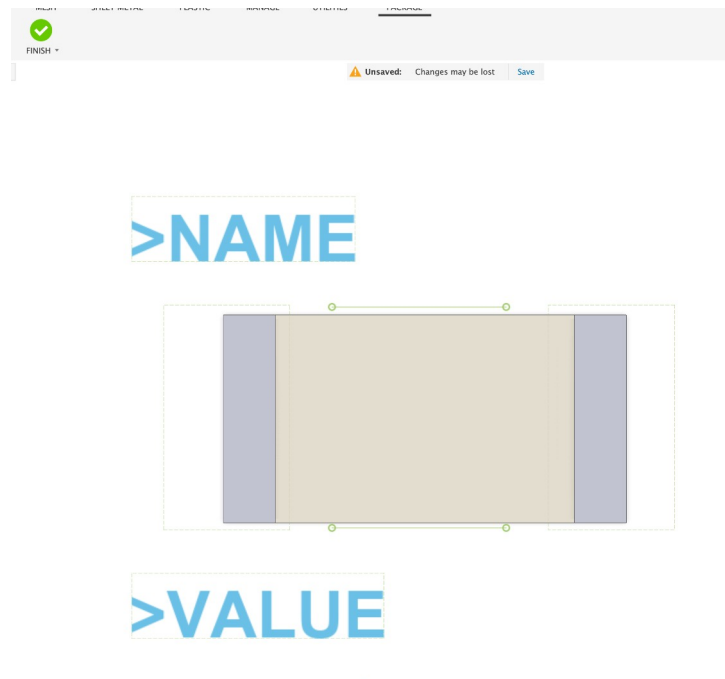


Figure 23

Next, click the Finish button (Green check mark icon in the tool bar).

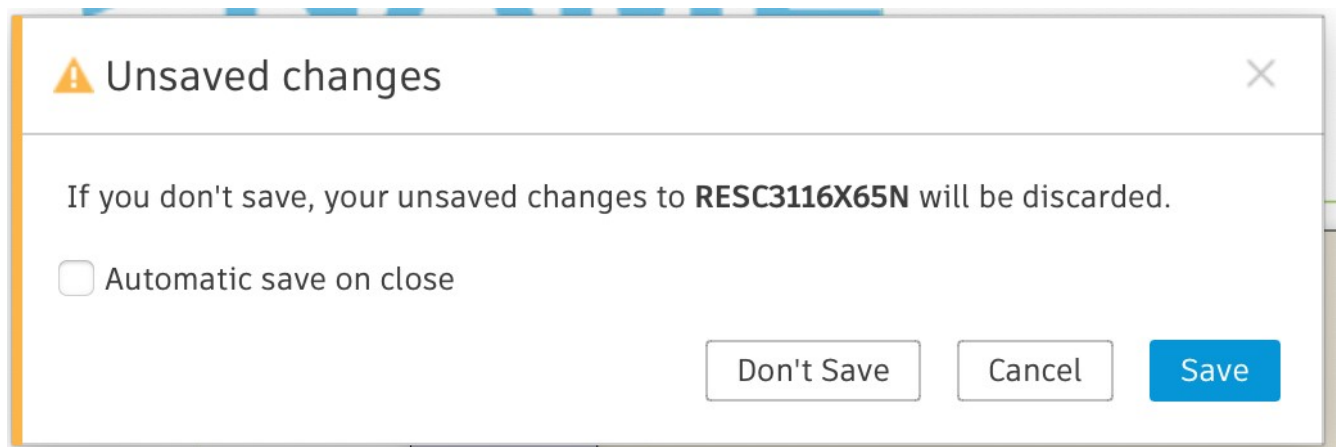


Figure 24

Ensure you are happy with the name and the location values then click the Save button.

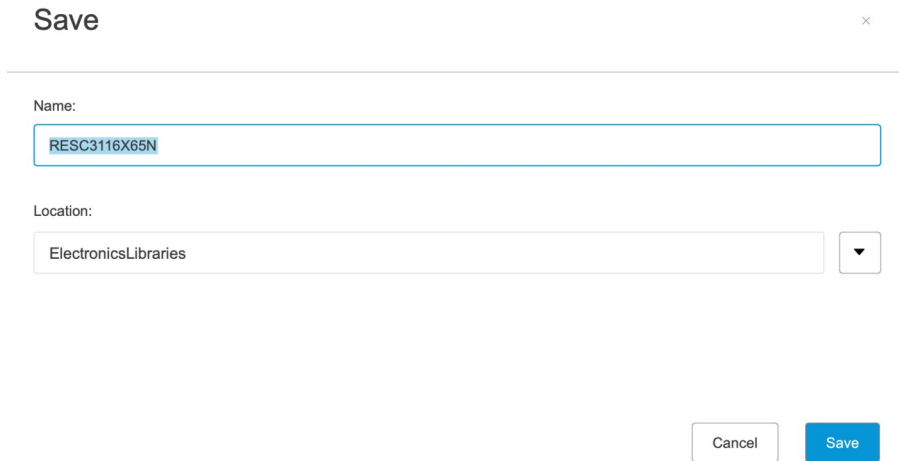


Figure 25

Now when you look at the Content manager pane on the left and look at the package associated with the footprint you will see the 3D model. Note that the 3D model does not yet appear in the right hand component property panel. This may be a bug with the Mac version of the software as of the writing of these notes.

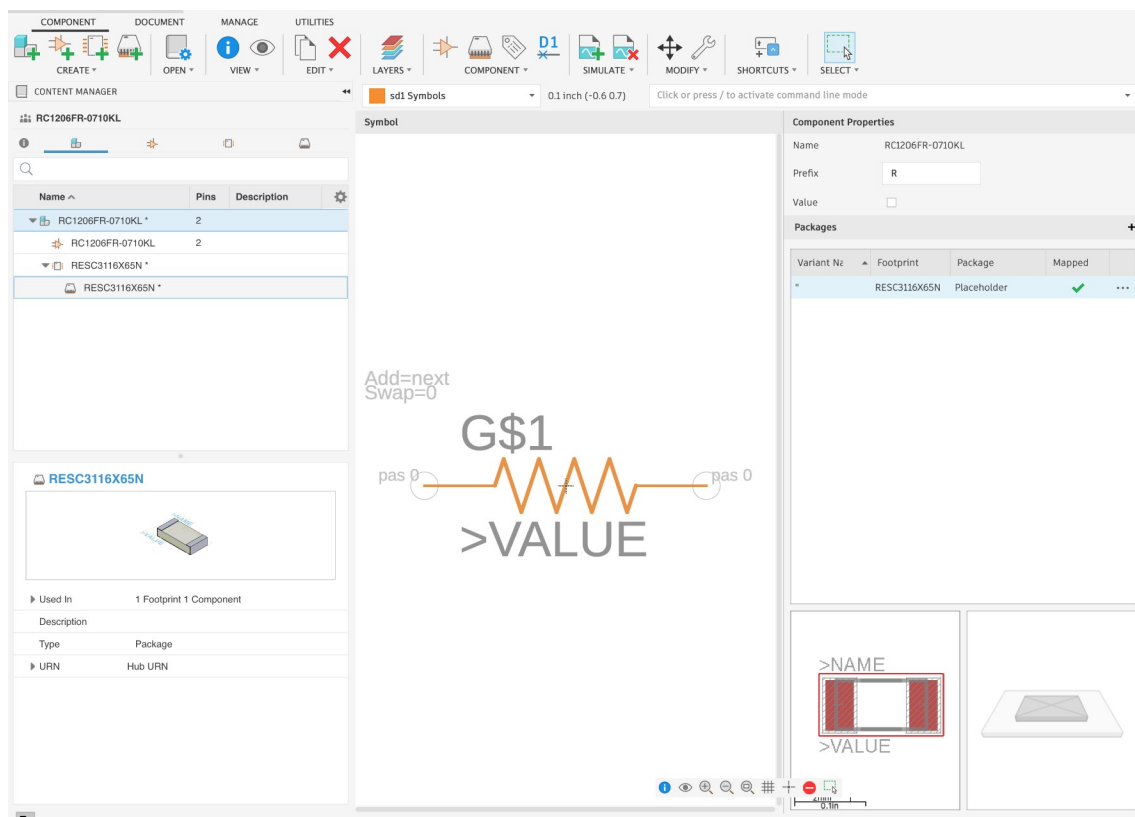


Figure 26

Click the save button in the application bar

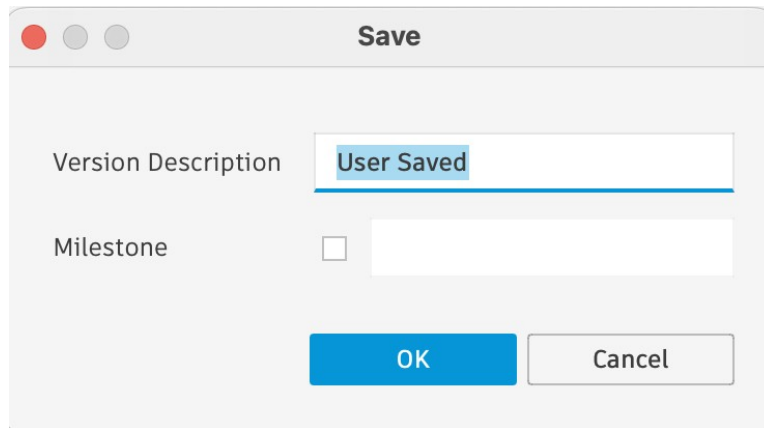


Figure 27

Add a description if you like (optional) then Click the OK button. When the file is done saving and uploading you can make the model appear in the component property panel by closing the component file then reopening it by double clicking it in the data panel (note this is NOT the package file that you just created but the original component file you used at the start of this process. You should now see the 3D model appear in both the Content Manager panel on the left under the package as well as in the right bottom corner of the component properties panel.

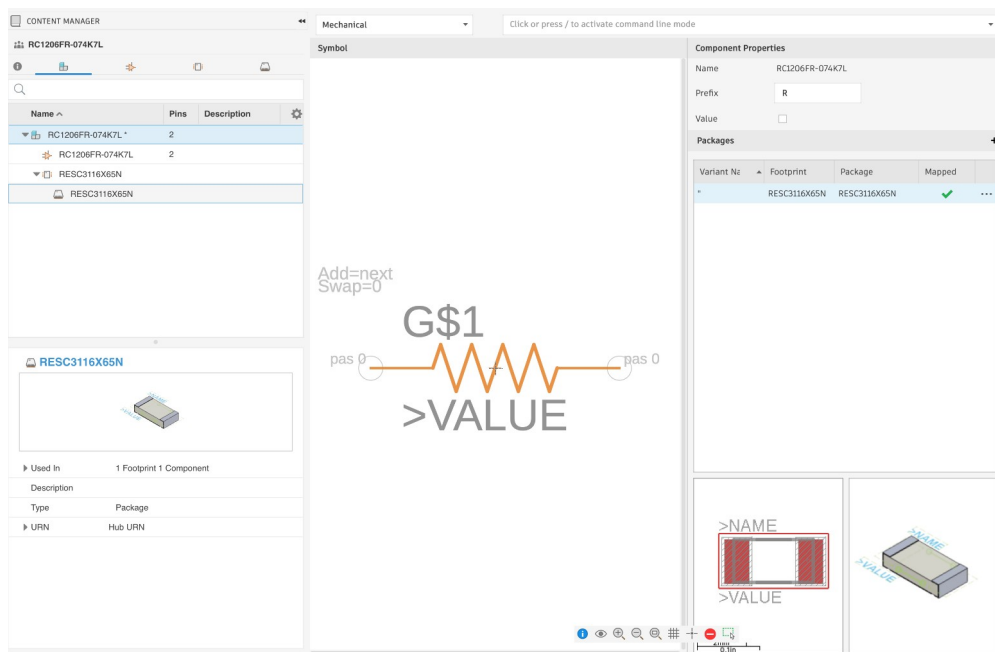


Figure 28

Congratulations, you have now created a component file in your electronics library project that contains the .lbr and .STEP assets that you got from Snap Magic. This library is referred to as you hub library.

Note that Fusion360 now considers the .lbr file you downloaded to your hard drive to be the local library. You should ignore your local libraries and use the hub library exclusively in your designs.